



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

AIML for Robotics

Module 1

Session Faculty:

Dr. Sangeetha Viswanathan



The course owner and the authors of this course deck, Prof. Raja vadhana and Dr.Sangeetha, are gratefully acknowledging all authors who made their course materials freely available online.

Course Content



☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

☐ Path Planning

- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Important Note : Hardware, physics, Mechanics, Dynamics, Kinematics, Sensors is not a part of this course. Application of Machine Learning & Deep Learning to the domain of Robotics for **MOBILE ROBOTS** with RIGID BODY is the **ONLY** focus of this course

Books



Text book

- Dieter Fox, Sebastian Thrun, Wolfram Burgard, "Probabilistic Robotics", Intelligent Robotics and Autonomous Agents series Ed, MIT Press, 2005
- Bruno Siciliano, Giuseppe Oriolo, Lorenzo Sciavicco, Luigi Villani, "Robotics: Modelling, Planning and Control", First Ed, Springer London Ltd, 2009

Reference book

- Ester Martinez Martin, "Swarm Robotics from Biology to Robotics", First Ed, IntechOpen, 2010
- Ramkumar Gandhinathan, "ROS Robotics Projects", Second Ed, Packt Publishing, 2019



Course Logistics



Evaluation Components and Schedule

	Evaluation Component	Mode	Duration	Weightage	Date and Time
EC1	Quiz I*	Online	15 Mins	5%	<u>35%</u> Refer to LMS
	Quiz II		15 Mins	5%	
	Assignment 01		4 Weeks	10%	
	Assignment 02 (+Situating Learning)		4 Weeks	15%	
EC2	Mid-sem exam (Closed book)		1.5-2 Hours	30%	
EC3	Comprehensive exam (Open book)		2.5 Hours	<u>35%</u>	

- Best of Two quizzes. Quiz may include programming questions at simple level as well.
- Strictly No Make up in EC1 components
- Late submission of Assignment will incur penalty . Check the LMS announcements for the details

Lab Sessions



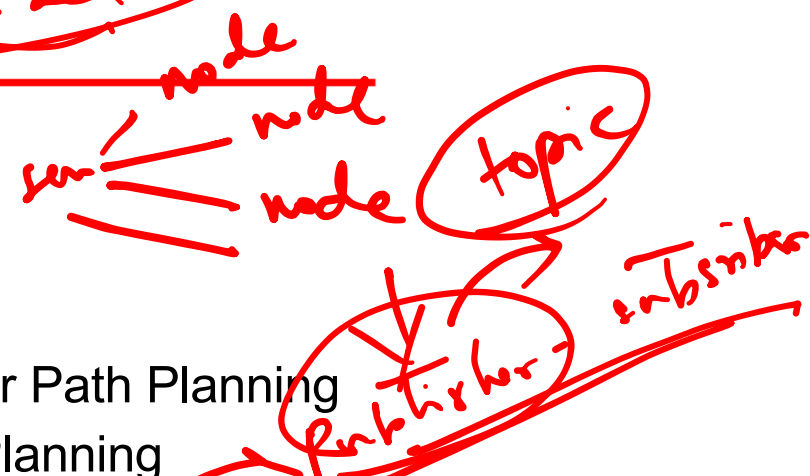
- L1 Introduction to Robotics Framework
- L2 Machine Learning for Robot Localization
- L3 CNN for Visual SLAM
- L4 Local Search(Evolutionary algorithms) for Path Planning
- L5 Deep Reinforcement Learning for Path Planning

Python Libraries

- PyBullet
- ROS-2 / Matlab (May be introduced)

Robot Operating System - 2

sens - ~~act~~ - set



Course Logistics



- Refer Canvas for the following
 - Handout
 - Schedule for Webinar
 - Schedule of Quiz, and Assignments.
 - Evaluation scheme
 - Session Slide Deck
 - Demo Lab Sheets
 - Quiz-I, Quiz-II
 - Assignment-I, Assignment-II
 - Review Questions
- Lecture Recordings
 - Available on Microsoft teams

Honour Code



All submissions for graded components must be the result of your original effort. It is strictly prohibited to copy and paste verbatim from any sources, whether online or from your peers. The use of unauthorized sources or materials, as well as collusion or unauthorized collaboration to gain an unfair advantage, is also strictly prohibited. Please note that we will not distinguish between the person sharing their resources and the one receiving them for plagiarism, and the consequences will apply to both parties equally.

In cases where suspicious circumstances arise, such as identical verbatim answers or a significant overlap of unreasonable similarities in a set of submissions, will be investigated, and severe punishments will be imposed on all those found guilty of plagiarism.

In case of Queries regarding the course....



Step 1: Post in the discussion forum.

- Read through the existing post and if you find any topic similar to your concern, add on to the existing discussion.
- Avoid duplication of queries or issues.

Step 2: Email to the IC at Prof. Sangeetha Viswanathan (sangeetha.viswanathan@pilani.bits-pilani.ac.in) if the query or issue is not resolved within 1 weeks time. Turn around for response to the email is 48hrs.

- In the subject pl mention the phrase "AIML for Robotics" clearly.
- Use BITS email id for correspondence. Emails from personal emails will be ignored without any reply.

PATIENCE is highly APPRECIATED

autonomous & autonomous
 — semi

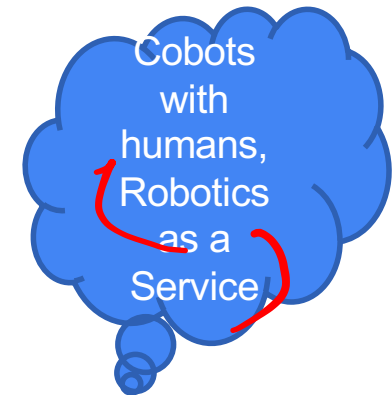
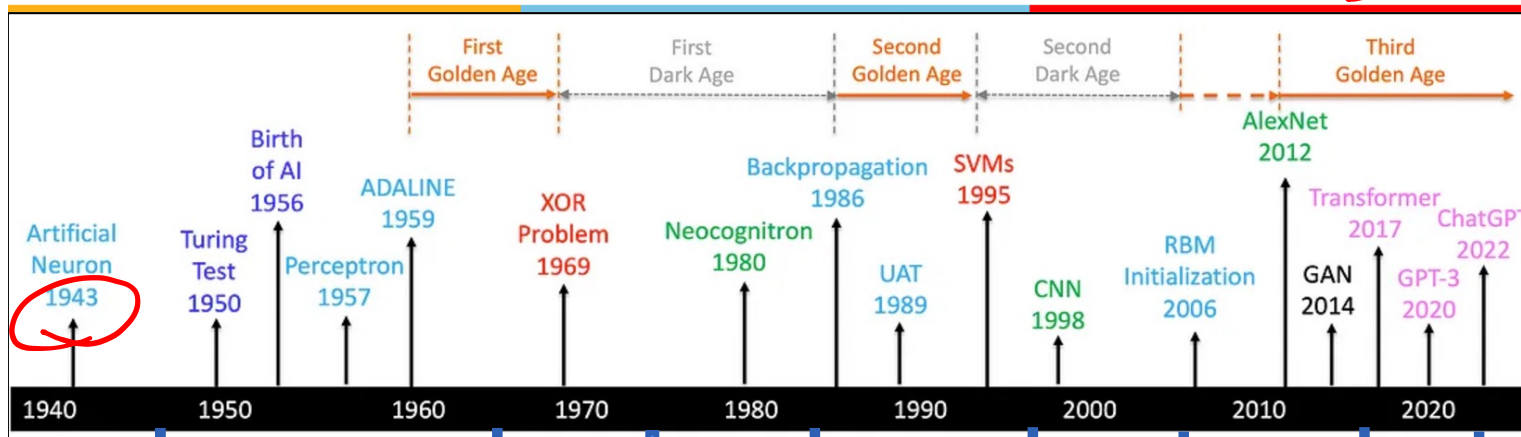
sense → reason → act

What is Robotics?

AI & ML

if x_1 → y_1
 if x_1 → y_2

Timeline : Intelligence in Robotics



Cybernetics
theory
(Norbert Wiener)
1947

AME



Shakey the Robot –
Stanford Research
Institute – 1966-69



Stanford Cart
Stanford AI Lab
1967-79

Robot Manufacturers
Copperweld Robotics
Unimation
Automatix, Inc.

Commercialization
with Vision tasks
1982-85

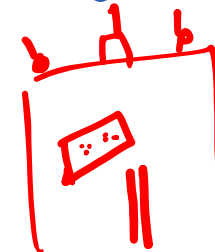
1990 First Autonomous
vehicles with real time
sensor proc
1996 Aibo by Sony
1997 - ROS



ROS
Self Driving Cars

ML/DNN
2015 - DRL- OpenAI
ROS adaption for Industrial
Robots
2014- Boston Dynamics Spot,
Atlas Robots

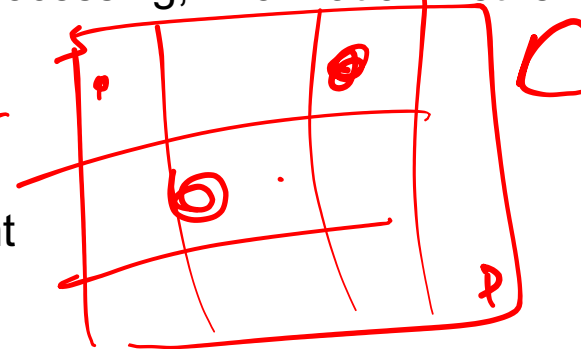
ROS2
5G, Edge Computing
Service Bots
RPA





Components of a Robotic System

- Actuation System
 - ✓ Locomotion
 - ✓ Manipulation
- Sensory System
 - ✓ Proprioceptive sensors. Eg., Position Sensors → internal state
 - ✓ Exteroceptive sensors. Eg., Cameras → external
 - ✓ Depend on : Signals, Data Processing, Information Retrieval
- Control System
 - ✓ Planning Techniques
 - ✓ Constraints from Agent
 - ✓ Constraints from Environment



Programmable Hardware Components

1. mp, embedded

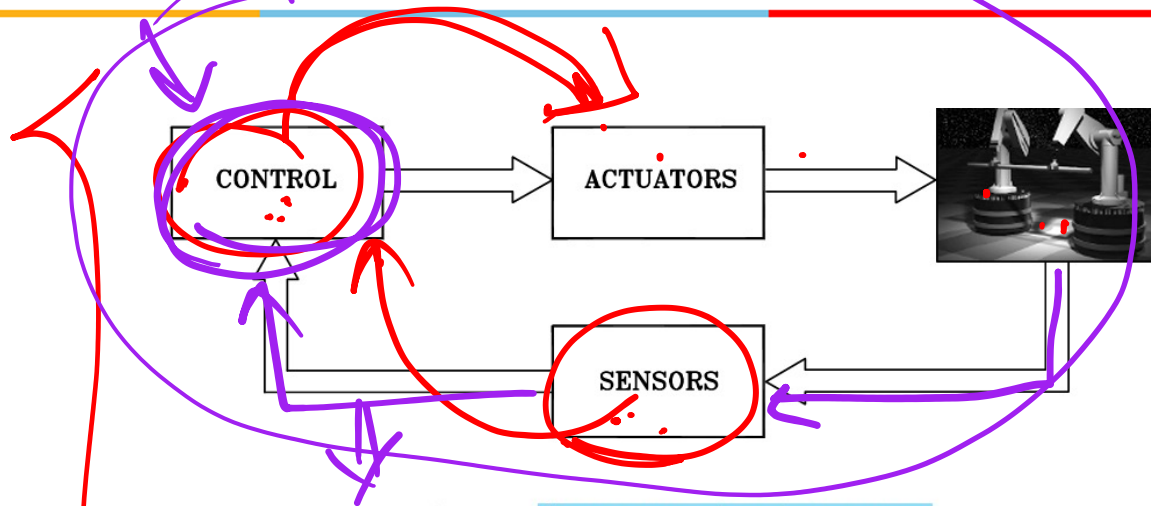


Fig. 1.1. Components of a robotic system

Physics
Mechanics
Kinematics
Dynamics

$S_1 A \rightarrow S_2$

kinematics
↓
 $X \xrightarrow{\text{dyn}} ?$

Dynamics
↓
 $X \xrightarrow{\text{load}} ?$



Types of Robots & Application

Terminologies



Degree of freedom (DoF)
 ↳ independent

Mobile Robots

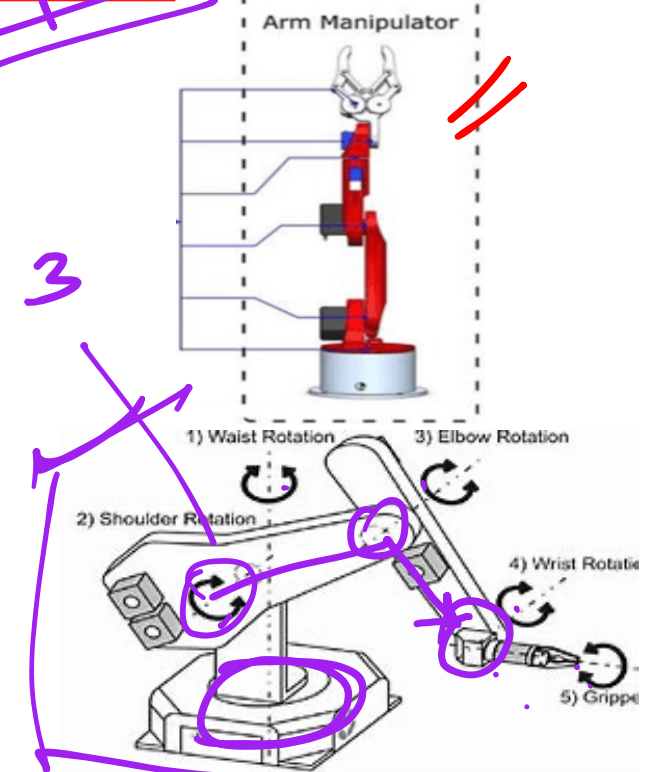
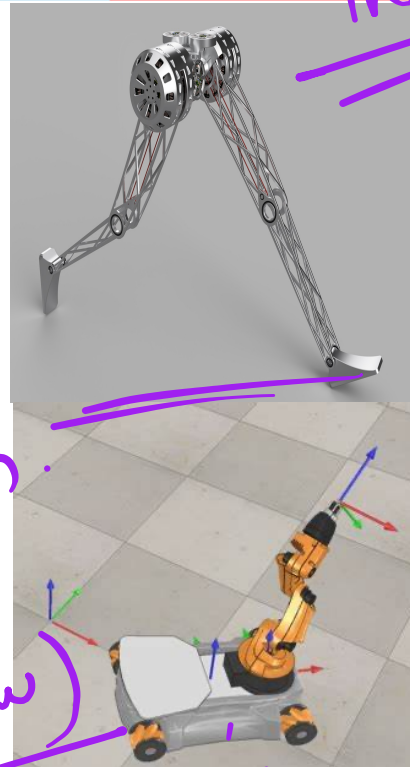
Manipulator Robots

- Frame/Body

pitch. ↕

mobile manipulator

hybrid → m + m
 6 DoF x, y, z → (roll, pitch, yaw)



Source Credit : [Robots : Degree of Freedom , Axis](#)

Terminologies

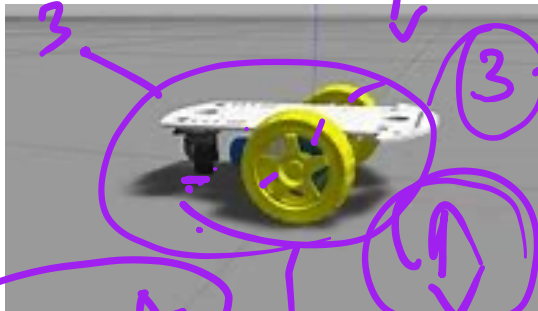
$b \rightarrow b$

$\updownarrow \rightarrow 1$

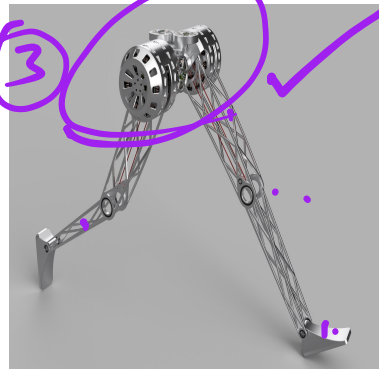


- Frame/Body
- Motor $\rightarrow$ actuator
- Fully/Under/Over Actuated

Fully: Differential Robot

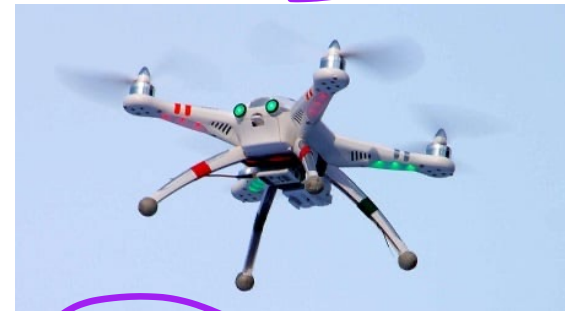


Under actuated: Bipedal



~~AKF~~
AKF

Over actuated : Fully autonomous



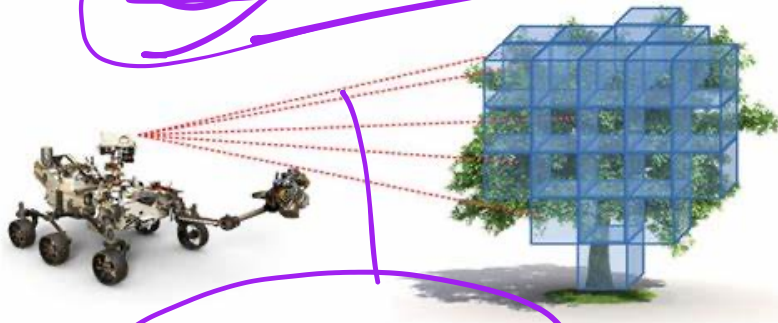
μ α

Terminologies



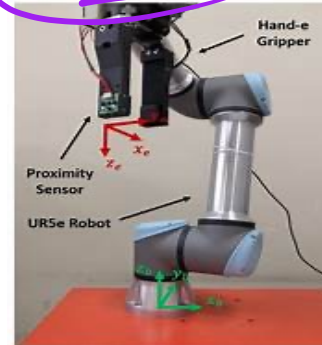
- Frame/Body
- Motor
 - Fully/Under/Over Actuated
- Sensor
 - Reactive/Sensor-based/Autonomous/Humanoid/**Mobile** etc.,

Vision



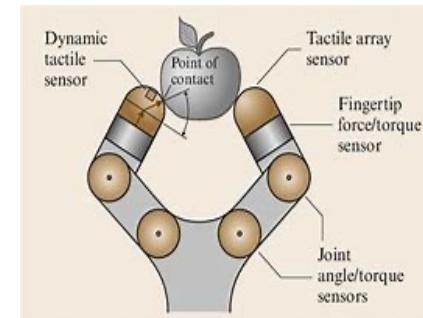
Autonomous Vehicle, Drones

Proximity

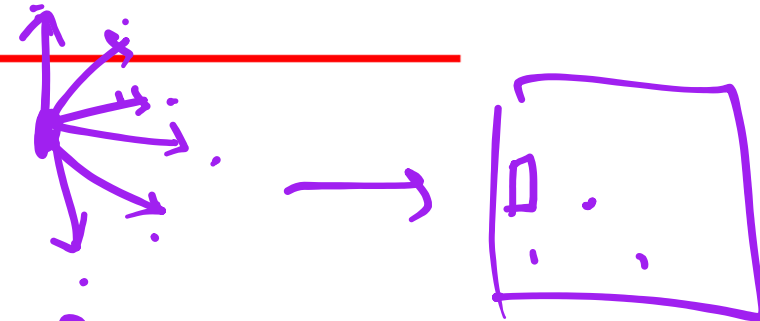


Vacuum Cleaner, Mobile Robots

Force Sensing



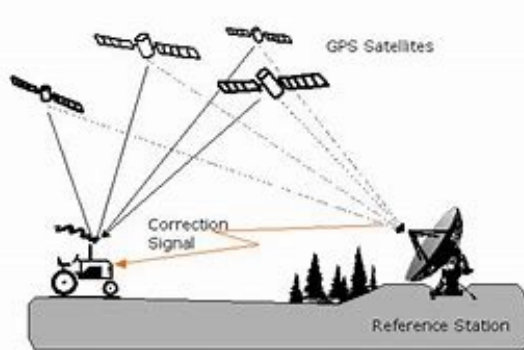
Surgical Robots



Terminologies

- Frame/Body
- Motor
 - Fully/Under/Over Actuated
- Sensor
 - ~~Reactive~~/Sensor-based/Autonomous/Humanoid/**Mobile** etc.,

Location



Warehouse bot, Autonomous Vehicle



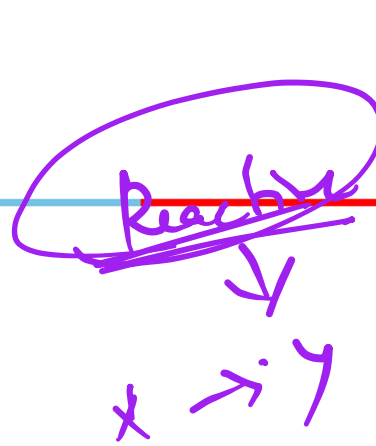
Acoustic (and many more.....)



Conversational Bots/Virtual Assistant



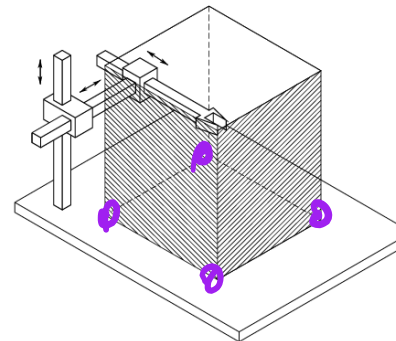
Multi Sensor Fusion Bots



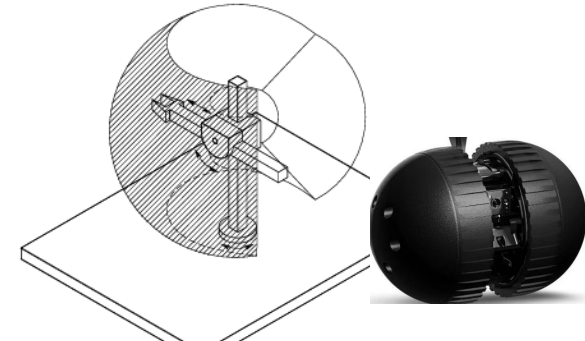
Terminologies



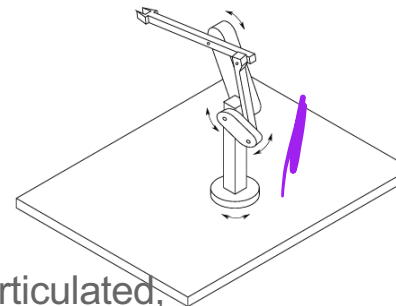
- Frame/Body
- Motor
 - Fully/Under/Over Actuated
- Sensor
 - Reactive/Sensor-based/Autonomous/Humanoid/**Mobile**
- Control System
 - Proportional-Integral-Derivative control (PID)
- Mechanics
 - Articulated/Cartesian/Spherical/Delta/**Wheeled(Tracked)**/Soft



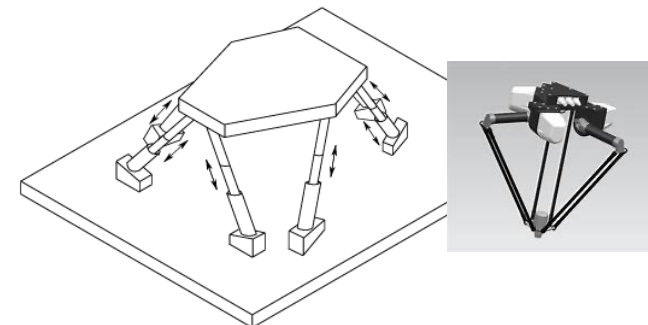
3D printers, laser-cutters



Surveillance, Exploration



Articulated,
With wheeled/legged
Nav tasks



High Speed Pick Place

Reference: T2 1.2.1 Overview only

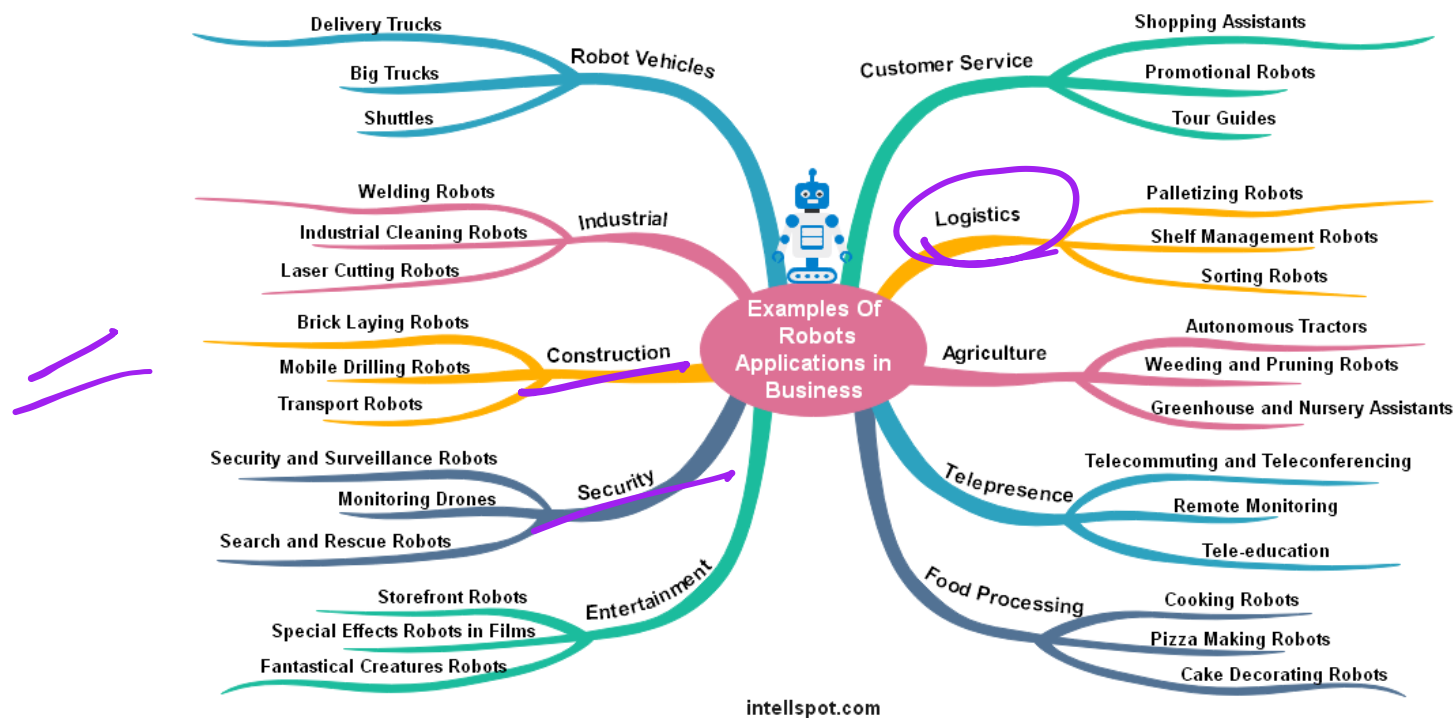
20

Terminologies



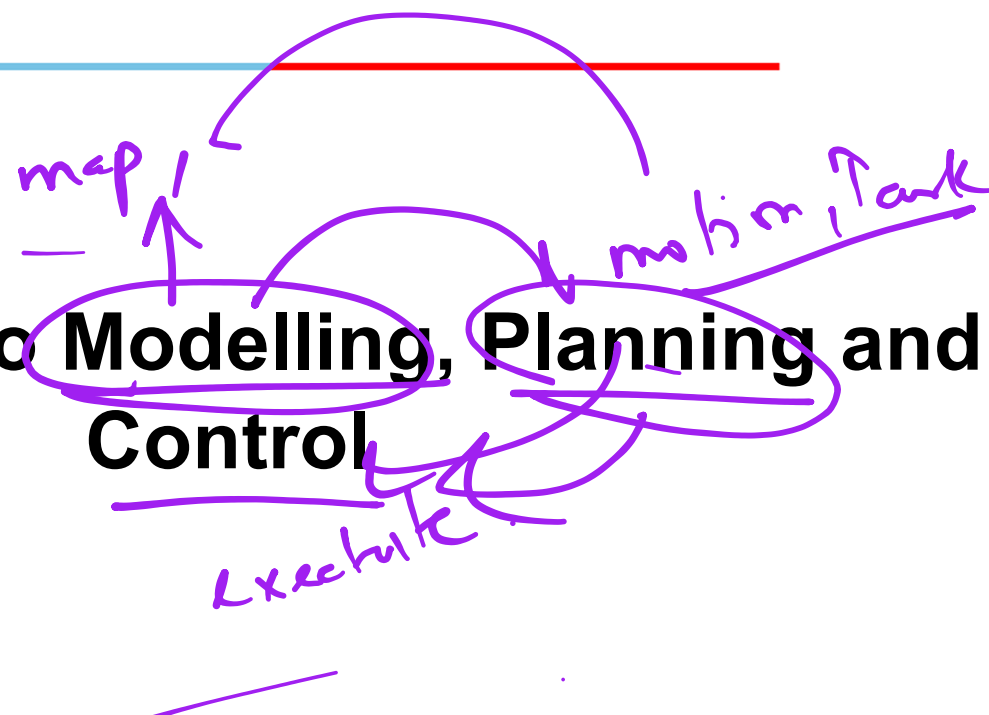
- Frame/Body
- Motor
 - Fully/Under/Over Actuated
- Sensor
 - Reactive/Sensor-based/Autonomous/Humanoid/**Mobile**
- Control System
 - Proportional-Integral-Derivative control (PID)
 - Learning
- Mechanics
 - Articulated/Cartesian/Spherical/Delta/**Wheeled(Tracked)**/Soft
- Dynamics
 - Underwater/Aerial/Swarm etc.,

Application Domain



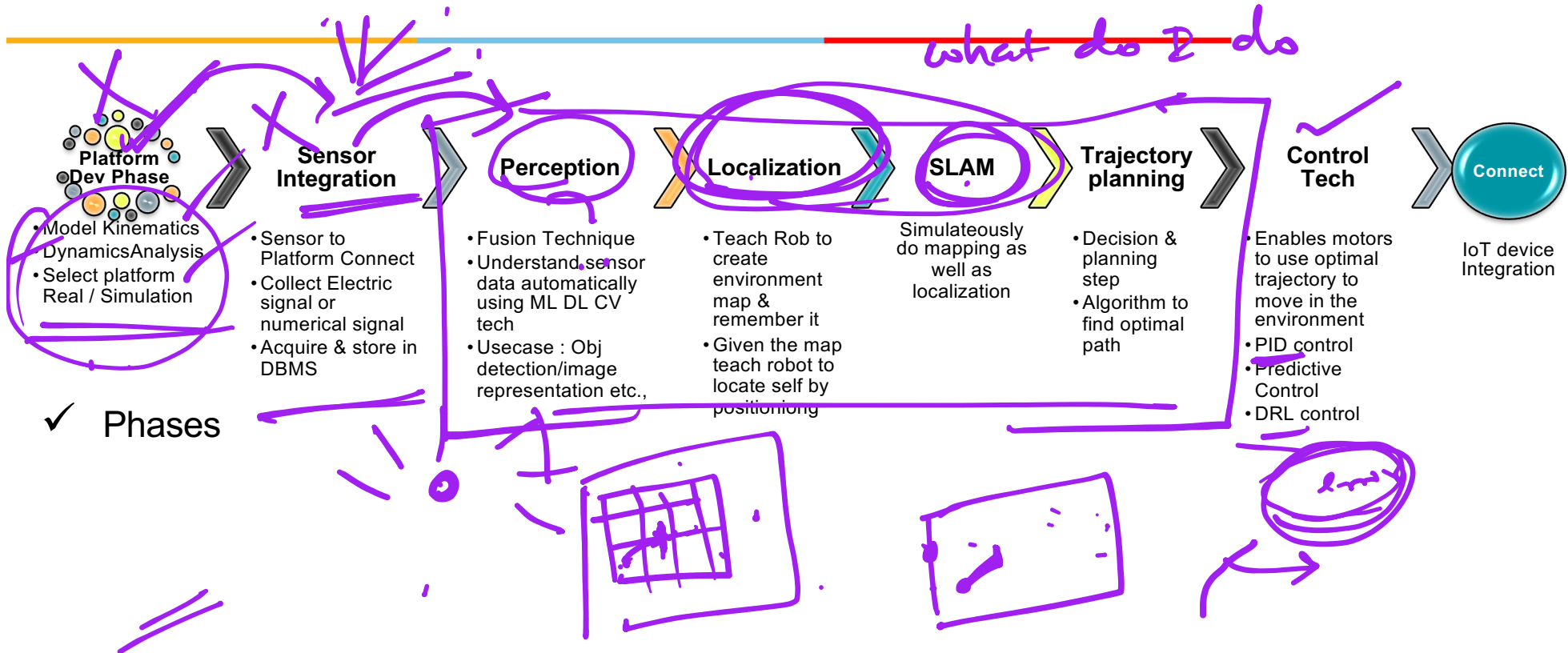
Source Credit : [10 Artificial Intelligence Robots Examples And Uses in Business](#)

Introduction to Modelling, Planning and Control



Phases of Robotic System Design

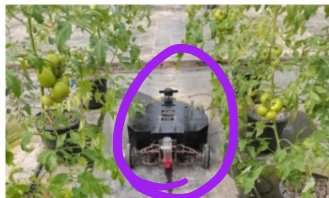
world like
where am I
what do I do



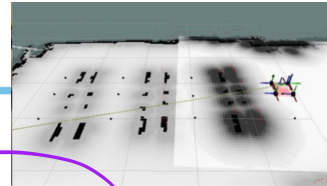
Case Study : Autonomous Navigation in Agriculture



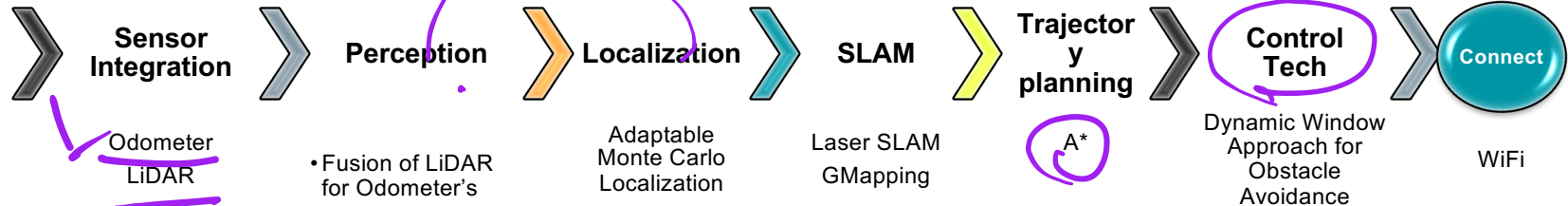
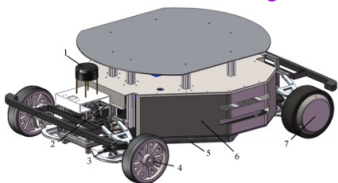
Objective	Auto Navigation of Greenhouse Robotic Agent for Harvesting
Domain	Agriculture
Scope of the Work	Focus on precision positional accuracy & variable path cruising Indoor



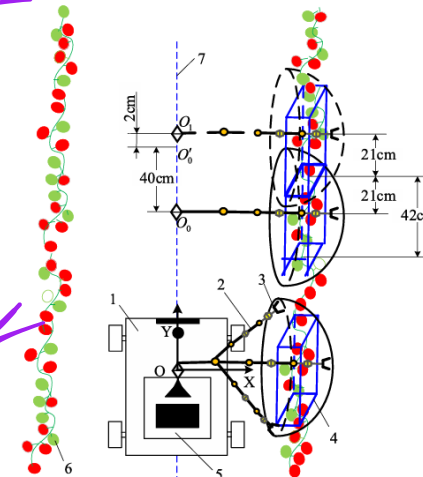
Case Study : 1



- Real PC monitor
- IPC
- LiDAR
- Motion control unit (STM32F103VET6)
- Two servo motors
- One steering engine
- An Encoder (OVW2-2MHT)
- ROS (Lunar Loggerhead)
- Ubuntu 16.04



Source Credit: Liu, K., Yu, J., Huang, Z., Liu, L., & Shi, Y. (2024). Autonomous navigation system for greenhouse tomato picking robots based on laser SLAM. *Alexandria engineering journal*, 100, 208-219.



Case Study : Autonomous Navigation in Agriculture



Objective	Auto Navigation of Greenhouse Robotic Agent for Harvesting
Domain	Agriculture
Scope of the Work	Focus on precision positional accuracy & variable path cruising Indoor
Key Area of Optimization	In the absence of good levelling construct map of the environment

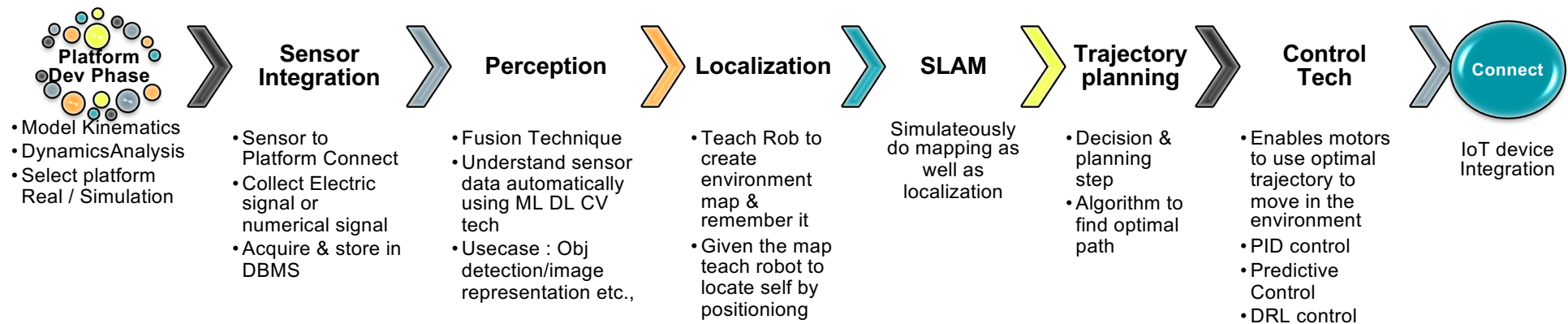


Case Study : Autonomous Navigation in Agriculture



Objective	Auto Navigation of Mobile Bots in unknown environment
Domain	*
Scope of the Work	Focus on DRL based path planning in the absence of MAP
Key Area of Optimization	In the absence of good levelling construct map of the environment

Case Study : 2



Source Credit: Ou, Y., Cai, Y., Sun, Y., & Qin, T. (2024). Autonomous Navigation by Mobile Robot with Sensor Fusion Based on Deep Reinforcement Learning. *Sensors*, 24(12), 3895. <https://doi.org/10.3390/s24123895>

Case Study : Cobots in Rescue Operation



Objective

Review of UAV Mapping, Co-localization and UGV Navigation in collaborative multi agent environment

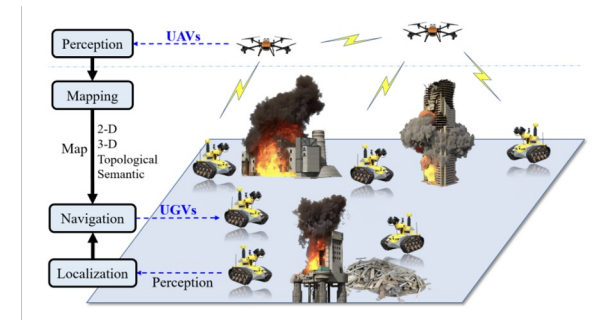
Domain

Fire & Rescue missions

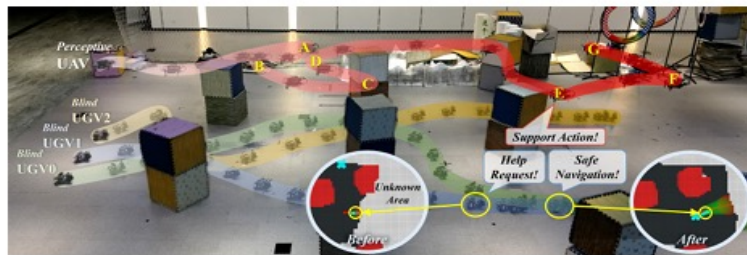
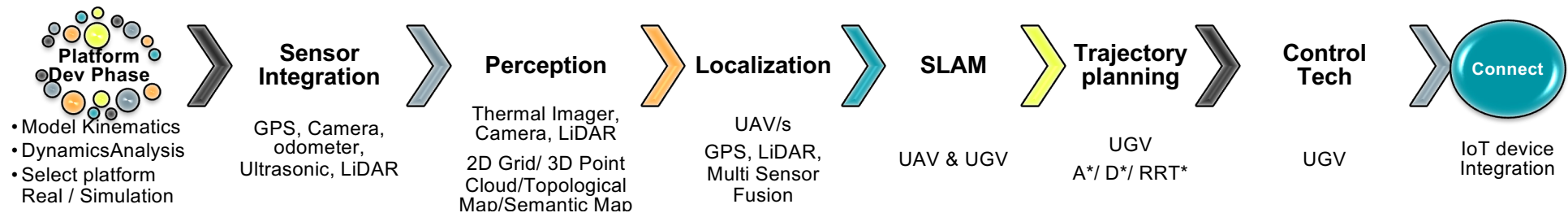
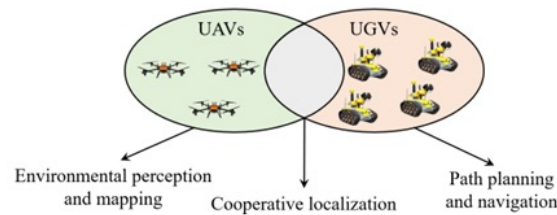
Scope of the Work

Focus on UGV using UAV's mapping for its navigation

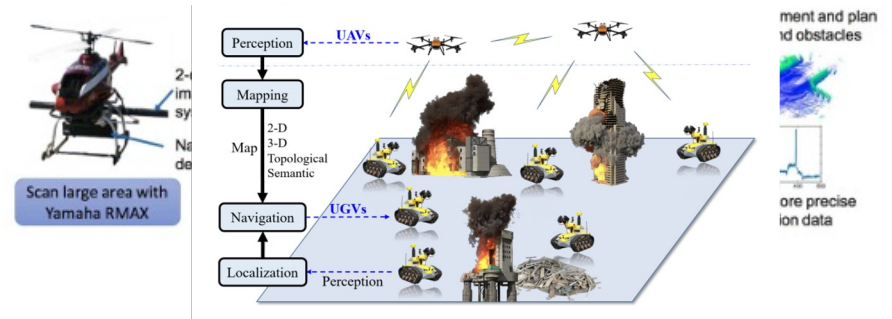
Source Credit: Zhang, Y., Yan, H., Zhu, D., Wang, J., Zhang, C. H., Ding, W., ... & Meng, M. Q. H. (2024). Air-Ground Collaborative Robots for Fire and Rescue Missions: Towards Mapping and Navigation Perspective. arXiv preprint arXiv:2412.20699.



Review



Source Credit: Zhang, Y., Yan, H., Zhu, D., Wang, J., Zhang, C. H., Ding, W., ... & Meng, M. Q. H. (2024). Air-Ground Collaborative Robots for Fire and Rescue Missions: Towards Mapping and Navigation Perspective. arXiv preprint arXiv:2412.20699.



Case Study : Cobots in Rescue Operation



Objective	Review of UAV Mapping, Co-localization and UGV Navigation in collaborative multi agent environment
Domain	Fire & Rescue missions
Scope of the Work	Focus on UGV using UAV's mapping for its navigation
Key Area of Optimization	Task Specific Feature Engineering, Multi modal sensor data Fusion, Learn new features in Real Time using Edge Computing, Embedded AI with Visual LLM for Scene Understanding

Source Credit: Zhang, Y., Yan, H., Zhu, D., Wang, J., Zhang, C. H., Ding, W., ... & Meng, M. Q. H. (2024). Air-Ground Collaborative Robots for Fire and Rescue Missions: Towards Mapping and Navigation Perspective. arXiv preprint arXiv:2412.20699.



Required Reading

- T1 Chapter 1
- R1 Chapter

Next Session Plan

- ✓ Introduction to Robot Motion Model
- ✓ Introduction to Robot Sensor Model

For the Next Class Prerequisite (Refer to the eLearning portal)